ORIGINAL ARTICLE

Lonvoguran Ziclumeran — In Vivo CRISPR Gene Editing in Hereditary Angioedema

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ABSTRACT

BACKGROUND
Hereditary angioedema is a rare and potentially life-threatening genetic condition characterized by recurrent and debilitating swelling attacks. Lonvoguran ziclumeran (lonvo-z) — an investigational, in vivo gene-editing treatment based on clustered regularly interspaced short palindromic repeats (CRISPR) — is being developed as a single-dose treatment for hereditary angioedema.

METHODS
In a phase 3, double-blind trial, we randomly assigned, in a 2:1 ratio, patients who were 16 years of age or older and had hereditary angioedema with C1 inhibitor deficiency to receive a single intravenous infusion of lonvo-z (at a dose of 50 mg) or placebo. The primary end point was the number of investigator-confirmed hereditary angioedema attacks per month (the monthly attack rate) from week 5 through week 28 after infusion.

RESULTS
A total of 80 patients underwent randomization; 52 were assigned to receive lonvo-z and 28 to receive placebo. As of February 10, 2026, the median duration of follow-up was 7.5 months (range, 4.9 to 12.8). The least-squares mean monthly attack rate from weeks 5 through 28 was 0.26 (95% confidence interval [CI], 0.15 to 0.45) in the lonvo-z group and 2.10 (95% CI, 1.55 to 2.86) in the placebo group (relative difference, −87%; 95% CI, −93 to −78; P<0.001). Adverse events that occurred during or after infusion of lonvo-z or placebo were reported in 92% and 86% of the patients in the lonvo-z and placebo groups, respectively, from day 1 through week 28. Among the adverse events that occurred during or after infusion and more often in the lonvo-z group than in the placebo group, the most common (occurring in >10% of the patients in the lonvo-z group) included infusion-related reaction, headache, fatigue, back pain, and upper respiratory tract infection. No serious or grade 3 or higher adverse events were reported in the lonvo-z group.

CONCLUSIONS
Among patients with hereditary angioedema, a single intravenous infusion of lonvo-z resulted in a significantly lower rate of hereditary angioedema attacks than placebo. (Funded by Intellia Therapeutics; HAELO ClinicalTrials.gov number, NCT06634420.)

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*A complete list of the HAELO investigators is provided in the Supplementary Appendix, available at NEJM.org.

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Hereditary angioedema is a rare, autosomal-dominant disorder characterized by unpredictable, recurrent, and potentially fatal attacks of swelling in the gastrointestinal tract, upper airway, and skin, which can result in substantial physical and psychosocial challenges.[1,2] Hereditary angioedema due to C1 inhibitor deficiency results from pathogenic variants in the gene that encodes C1 inhibitor (*SERPING1*), which causes either low C1 inhibitor levels or impaired C1 inhibitor function.[2] This deficiency results in dysregulated plasma kallikrein activity and excessive bradykinin production, which lead to episodes of increased vascular permeability and swelling.[2]

Hereditary angioedema remains a challenge for patients despite the availability of on-demand therapy for the treatment of attacks and long-term prophylactic medications. Although the availability of on-demand therapy and long-term prophylaxis has improved the management of hereditary angioedema, incomplete disease control, lack of access to treatments, and the need for lifelong use still impose a burden on patients. Results of phase 3 trials of long-term prophylaxis suggest that approximately 40 to 60% of patients can become attack-free.[3-5] However, in a 2025 survey of patients with hereditary angioedema who were not participating in clinical trials, 80% of the patients reported having attacks, even though 89% of the patients reported using long-term prophylaxis.[6] Fear and anxiety about the next attack profoundly affect patients, and this psychosocial burden is influenced by the degree of attack control.[7]

Reducing plasma kallikrein production or activity is a clinically validated treatment strategy to reduce bradykinin production, although currently approved treatment options require chronic or on-demand dosing to show efficacy.[2-5,8,9] Plasma kallikrein is derived from prekallikrein, and since the liver is the primary source of prekallikrein,[10] the use of a nonviral, liver-directed lipid nanoparticle to deliver clustered regularly interspaced short palindromic repeats (CRISPR) gene editing in vivo is being investigated as a therapeutic strategy. Lonvoguran ziclumeran (also known as lonvo-z or NTLA-2002) contains CRISPR components — a guide RNA that targets the prekallikrein gene *KLKB1* and a codon-optimized human mRNA sequence of *Streptococcus pyogenes* Cas9 protein. Lonvo-z was designed to reduce the level of plasma prekallikrein by inactivating *KLKB1*, with the intent of resetting the balance of the kallikrein–kinin system and normalizing bradykinin levels.[10]

In a phase 1–2 trial (ClinicalTrials.gov number, NCT05120830), all 37 of the patients who received treatment with lonvo-z had rapid and deep reductions in total plasma kallikrein levels that were sustained through the last follow-up visit; 20 of the 21 patients treated with 50 mg of lonvo-z became attack-free without the use of long-term prophylaxis.[11] The optimal biologic dose of lonvo-z was identified as 50 mg. Here we report the primary results from HAELO (A Phase 3 Study to Evaluate NTLA-2002 in Participants with Hereditary Angioedema), which evaluated the efficacy and safety of a single intravenous infusion of 50 mg of lonvo-z in patients with hereditary angioedema due to C1 inhibitor deficiency (Fig. S1).

Methods

Trial Design and Oversight

We conducted this multinational, double-blind, randomized, placebo-controlled trial at 31 sites in Australia, Canada, France, Germany, the Netherlands, New Zealand, South Africa, the United Kingdom, and the United States. The trial was approved by the institutional review board or ethics committee at each participating trial site. All the patients provided written informed consent. The trial was conducted in accordance with the principles of the Declaration of Helsinki and the Good Clinical Practice guidelines of the International Council for Harmonisation. An independent data monitoring committee assessed trial conduct and interim safety summaries during the trial. The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol, available with the full text of this article at NEJM.org. The first draft of the manuscript was written by medical writers funded by Intellia Therapeutics. The authors reviewed and edited the manuscript and made the decision to submit it for publication. The contributions of individual authors are provided in the Supplementary Appendix (available at NEJM.org).

Patients

Eligible patients were at least 16 years of age, had received a diagnosis of hereditary angioedema due to C1 inhibitor deficiency, and had at least two investigator-confirmed hereditary angioedema attacks during the 4- to 8-week run-in period. Patients were not permitted to use long-term prophylaxis within five half-lives before the start of the run-in period through the end of week 28. Full eligibility criteria are listed in the protocol. Screening was stopped when it was estimated that a sufficient number of patients would be enrolled.

RANDOMIZATION AND TREATMENT

Patients were randomly assigned in a 2:1 ratio to receive a single infusion of lonvo-z at a dose of 50 mg or placebo. The baseline number of attacks per month (≤3 or >3) was used as a stratification factor for randomization. On day 1, the infusion was administered intravenously over the course of approximately 4 hours. The efficacy evaluation period was defined as week 5 (day 29) through week 28 (day 197) after infusion. During this time, no use of long-term prophylaxis was permitted. At week 28, patients in both trial groups could elect to cross over to receive a single blinded infusion of placebo (for patients in the lonvo-z group) or lonvo-z (for patients in the placebo group). All patients will undergo additional observation for 76 weeks after the end of the efficacy evaluation period, for a total trial duration of 104 weeks.

To mitigate the risk of infusion-related reactions, patients received an oral glucocorticoid (dexamethasone at a dose of 8 mg or equivalent therapy) at home 8 to 24 hours before infusion. In addition, 1 to 2 hours before infusion, patients received either an oral or intravenous glucocorticoid (dexamethasone [10 mg] or equivalent therapy) and H_1 and H_2-antihistamines: intravenous diphenhydramine (50 mg or equivalent) or oral cetirizine (10 mg or equivalent) and intravenous or oral famotidine (20 mg or equivalent). To mitigate the risk of a hereditary angioedema attack during the infusion (given the washout of long-term prophylaxis), C1 inhibitor short-term prophylaxis was required immediately before infusion. Access to therapy to treat attacks was required throughout the trial.

CLINICAL ASSESSMENT

Each patient recorded angioedema symptoms, attacks, and treatments in an electronic diary that was reviewed by an investigator to confirm whether an event represented a hereditary angioedema attack. Blood samples for pharmacodynamic and pharmacokinetic analyses were obtained. Additional evaluations are described in the protocol.

TRIAL END POINTS

The primary end point was the time-normalized number of investigator-confirmed hereditary angioedema attacks per month (the attack rate) from week 5 through week 28, with a month defined as 28 days. Key secondary end points from week 5 through week 28 were the time-normalized number of investigator-confirmed hereditary angioedema attacks per month requiring on-demand therapy, the time-normalized number of moderate or severe investigator-confirmed attacks per month, and investigator-confirmed attack-free status. The change from baseline (last assessment before randomization) to week 28 in the total score on the Angioedema Quality of Life (AE-QoL) questionnaire was also a key secondary end point. The AE-QoL questionnaire is a validated, angioedema-specific patient-reported outcome measure; total scores range from 0 to 100, with lower scores indicating less impairment.

Other secondary end points were response status (a reduction from baseline of at least 50%, 70%, or 90% in the time-normalized number of investigator-confirmed hereditary angioedema attacks per month from week 5 through week 28) and the time-normalized number of investigator-confirmed attacks per month from week 1 through week 28. Additional secondary end points from week 1 through week 28 included the number of attacks per month requiring on-demand therapy, the number of moderate or severe attacks, attack-free status, and response status. End points related to long-term efficacy at week 104 are described in the protocol. Additional end points included safety (as assessed on the basis of adverse events), pharmacodynamics (the change from baseline in total plasma kallikrein protein level), pharmacokinetics, and immunogenicity.

STATISTICAL ANALYSIS

We planned to enroll a total of 60 patients (40 patients in the lonvo-z group and 20 patients in the placebo group), which would provide the trial with 99% power to detect a monthly attack rate ratio of 0.25 for the difference between the lonvo-z group and the placebo group, with a two-sided alpha level of 0.05.

Formal statistical hypothesis testing was performed for the primary end point and four key secondary end points with a hierarchical gatekeeping testing procedure. If a significant difference was observed for the primary end point, the key secondary end points were to be evaluated in the following hierarchical order: the time-normalized number of investigator-confirmed hereditary angioedema attacks requiring on-demand therapy from week 5 through week 28, the time-normalized number of moderate or severe investigator-confirmed attacks from week 5 through week 28, investigator-confirmed attack-free status from week 5 through week 28, and the change from baseline to week 28 in the AE-QoL questionnaire total score. The secondary end points were tested in the order described only if the primary end point and each preceding secondary end point in the hierarchy were statistically significant. The family-wise type I error rate was controlled at a two-sided alpha level of 0.05. The primary analysis was based on a prespecified completion date after at least 60 patients had completed the week 28 visit or had discontinued the trial.

From week 5 through week 28, the number of attacks, attacks requiring on-demand therapy, and moderate or severe attacks were analyzed with a Poisson regression model with Pearson chi-square scaling of standard errors to account for potential overdispersion. The model included trial group and baseline attack rate (≤3 or >3 attacks per month) as covariates, and the logarithm of follow-up duration for each patient from week 5 through week 28 as the offset variable. Attack-free status from week 5 through week 28 was assessed with the Cochran–Mantel–Haenszel test, with stratification according to the baseline attack rate. The change from baseline to week 28 in AE-QoL questionnaire total score was assessed with a mixed model for repeated measures that included trial group, AE-QoL questionnaire total score at baseline, visit, and trial-group-by-visit interaction as independent variables.

For patients who started long-term prophylaxis before the week 28 assessment, only attacks that started before initiation of long-term prophylaxis were included in the analysis. For patients who had not yet completed the week 28 visit at the time of the data-cutoff date (February 10, 2026), all data through their latest attack assessment were included in the analysis. Missing data were assumed to be missing at random since the unobserved data were administratively censored and no imputation was performed in the analysis. In a sensitivity analysis, multiple imputation was performed under missing-at-random and jump-to-reference assumptions for the primary and key secondary end points. Additional statistical methods are provided in the Supplementary Appendix. The 95% confidence intervals have not been not adjusted for multiplicity and should not be used in place of hypothesis testing.

The full analysis population, which was used for efficacy analyses, included all the patients who had undergone randomization. The safety analysis population included all the patients who had undergone randomization and received at least a partial dose of lonvo-z or placebo. Safety data were summarized with the use of descriptive statistics according to trial group.

RESULTS

PATIENTS

Owing to an unexpectedly low incidence of screening failure, 80 of the 92 screened patients underwent randomization; 52 patients were assigned to receive lonvo-z and 28 were assigned to receive placebo (Fig. S2). As of February 10, 2026 (the data-cutoff date), the median follow-up was 7.5 months (range, 4.9 to 12.8). The data from the primary analysis are reported here, and longer-term follow-up of the trial is ongoing. A total of 64 patients (43 in the lonvo-z group and 21 in the placebo group) reached at least the week 28 evaluation; the remaining 16 patients reached at least week 20. As of the data-cutoff date, all 80 patients continued to be followed and were included in the primary analysis.

Forty-nine patients received the full dose of lonvo-z. Two patients received a nearly full dose (97% and 98% of the planned dose), with nominal volume loss related to the management of occlusions during infusion. One patient received a partial dose (17% of the planned dose) due to an infusion pump error that resulted in the infusion duration exceeding the permitted time frame.

The demographic and clinical characteristics of the patients at baseline were balanced in the two groups (Table 1 and Table S1) and were generally representative of the population of patients with hereditary angioedema (Table S2). The median age of the patients was 41.5 years (range, 19.0 to 76.0), and 71% of the patients were receiving long-term prophylaxis before enrollment. The

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

Characteristic	Lonvo-z (N=52)	Placebo (N=28)	All Patients (N=80)
Median age at screening (range) — yr	42.0 (23.0–71.0)	40.0 (19.0–76.0)	41.5 (19.0–76.0)
Sex — no. (%)			
Female	35 (67)	20 (71)	55 (69)
Male	17 (33)	8 (29)	25 (31)
Type of hereditary angioedema — no. (%)†			
C1INH type 1	49 (94)	25 (89)	74 (92)
C1INH type 2	3 (6)	3 (11)	6 (8)
Previous treatment for hereditary angioedema before trial enrollment — no. (%)			
Long-term prophylaxis‡	35 (67)	22 (79)	57 (71)
On-demand treatment only	17 (33)	6 (21)	23 (29)
Typical severity of angioedema attacks — no. (%)			
Mild	7 (13)	5 (18)	12 (15)
Moderate	30 (58)	20 (71)	50 (62)
Severe	15 (29)	3 (11)	18 (22)
No. of hereditary angioedema attacks per month during the run-in period	3.5±1.79	3.5±1.91	3.5±1.82
AE-QoL total score§	58.4±18.05	58.0±12.54	58.3±16.25

* Complete data on demographic and clinical characteristics are provided in the Supplementary Appendix. Plus–minus values are means ±SD. Percentages may not total 100 because of rounding.

† Hereditary angioedema C1 inhibitor (C1INH) type 1 includes patients with low C1 inhibitor levels; C1INH type 2 includes those with impaired C1 inhibitor function.

‡ Long-term prophylaxis before trial enrollment was defined as the last dose of long-term prophylaxis being discontinued within 100 days before the first screening visit.

§ The Angioedema-Quality of Life (AE-QoL) questionnaire is a validated, angioedema-specific patient-reported outcome measure; total scores range from 0 to 100, with lower scores indicating less impairment.

mean (±SD) baseline monthly attack rate during the run-in period was 3.5±1.8 in the lonvo-z group and 3.5±1.9 in the placebo group.

EFFICACY

Primary End Point

The least-squares mean monthly rate of investigator-confirmed hereditary angioedema attacks from week 5 through week 28 was 0.26 (95% confidence interval [CI], 0.15 to 0.45) in the lonvo-z group and 2.10 (95% CI, 1.55 to 2.86) in the placebo group (relative difference, −87%; 95% CI, −93 to −78; P<0.001) (Table 2). The difference in the mean number of attacks per month between the two trial groups appeared within 4 weeks after infusion, and the response to lonvo-z remained stable through follow-up, albeit in diminishing numbers of patients at later time points (Fig. 1). The responses of individual patients are depicted in Figure S3. The results appeared to be generally consistent in prespecified subgroup analyses (Fig. S4).

Key Secondary End Points

The least-squares mean monthly rate of investigator-confirmed hereditary angioedema attacks requiring on-demand therapy from week 5 through week 28 was 0.19 (95% CI, 0.10 to 0.36) in the lonvo-z group and 1.79 (95% CI, 1.27 to 2.54) in the placebo group (relative difference, −89%; 95% CI, −94 to −79; P<0.001). The least-squares mean monthly rate of moderate or severe attacks from week 5 through week 28 was 0.11 (95% CI, 0.06 to 0.23) in the lonvo-z group and 1.23 (95% CI, 0.84 to 1.81) in the placebo group (relative difference, −91%; 95% CI, −96 to −81; P<0.001). From week 5 through week 28, the patients who remained attack-free included 32 of 52 patients (62%; 95% CI, 47 to 75) in the lonvo-z group and 3 of 28 patients (11%; 95% CI, 2 to 28) in the

Table 2. Primary and Key Secondary End Points.

End Point	Lonvo-z (N=52)	Placebo (N=28)
Primary end point		
LS mean no. of investigator-confirmed hereditary angioedema attacks per month, wk 5 through wk 28 (95% CI)*†‡	0.26 (0.15 to 0.45)	2.10 (1.55 to 2.86)
Percent difference vs. placebo (95% CI)	−87 (−93 to −78)	
P value	<0.001	
Key secondary end points		
LS mean no. of investigator-confirmed hereditary angioedema attacks requiring on-demand therapy, wk 5 through wk 28 (95% CI)*†‡	0.19 (0.10 to 0.36)	1.79 (1.27 to 2.54)
Percent difference vs. placebo (95% CI)	−89 (−94 to −79)	
P value	<0.001	
LS mean no. of moderate or severe hereditary angioedema attacks, wk 5 through wk 28 (95% CI)*†‡	0.11 (0.06 to 0.23)	1.23 (0.84 to 1.81)
Percent difference vs. placebo (95% CI)	−91 (−96 to −81)	
P value	<0.001	
Attack-free status, wk 5 through wk 28 — no. (%)†‡	32 (62)	3 (11)
95% CI§	47.0 to 74.7	2.3 to 28.2
Odds ratio vs. placebo (95% CI)	12.81 (3.45 to 47.55)	
P value	<0.001	
LS mean change from baseline to wk 28 in AE-QoL questionnaire total score — (95% CI)‡¶	−23.51 (−27.64 to −19.38)	−6.47 (−12.26 to −0.68)
LS mean difference vs. placebo (95% CI)	−17.04 (−24.15 to −9.93)	
P value	<0.001	

* The mean number of hereditary angioedema attacks per month was estimated with the use of a Poisson regression model with Pearson chi-square scaling of standard error with treatment group and baseline attack rate (≤3 or >3 attacks per month) as covariates. Baseline was defined as the time from the date of the second screening visit to randomization. A month was defined as 28 days.

† For patients who initiated any long-term prophylaxis before the week 28 assessment, only attacks that started before prophylaxis initiation were included in the analysis. For patients who did not reach week 28 assessment, all data through their latest attack assessment were included.

‡ All 16 patients who had not yet reached week 28 at the time of the data-cutoff date owing to later enrollment remained in follow-up, and all available data through their latest assessment were included in the analysis for the primary and key secondary end points. No imputation was performed in the analyses.

§ The exact confidence intervals were calculated with the Clopper–Pearson method.

¶ The AE-QoL questionnaire total score ranges from 0 to 100, with lower scores indicating better quality of life. The minimal clinically important difference is defined as a change of 6 points.[12] The total score change from baseline at week 28 was estimated with the use of a mixed model for repeated measures, with baseline total score, trial group, visit, and trial-group-by-visit interaction as fixed effects, and with patient as a random effect. For patients who initiated any long-term prophylaxis before the week 28 assessment, only AE-QoL data that were obtained before initiation of prophylaxis were included in the analysis. At week 28, data were available for 44 patients in the lonvo-z group and 20 patients in the placebo group.

placebo group (odds ratio, 12.81; 3.45 to 47.55; P<0.001). Patients in the lonvo-z group had a greater improvement in the AE-QoL total score from baseline to week 28 than those in the placebo group, with a least-squares mean change from baseline of −23.51 points (95% CI, −27.64 to −19.38) in the lonvo-z group and −6.47 points (95% CI, −12.26 to −0.68) in the placebo group (mean difference, −17.04 points; 95% CI, −24.15 to −9.93; P<0.001), a finding that exceeded the minimal clinically important difference of a change of 6 points in the AE-QoL total score.[12]

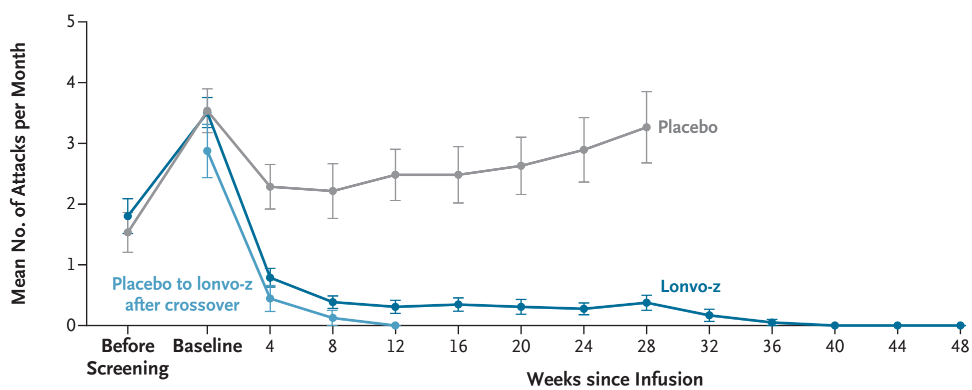
Other End Points

From week 5 through week 28, a reduction from baseline of 90% or more in the monthly attack rate was observed in 39 of the 52 patients (75%) in the lonvo-z group and in 5 of the 28 patients (18%) in the placebo group (Fig. S5A). Similar results were observed from week 1 through week 28 (Fig. S5B). From week 1 through week 28, the least-squares mean monthly attack rate was 0.32 (95% CI, 0.20 to 0.51) in the lonvo-z group and 2.07 (95% CI, 1.55 to 2.75) in the placebo group (relative difference, −84%; 95% CI, −91 to −75) (Table S3). From week 5 through the last follow-up assessment, among patients treated with lonvo-z, the mean reduction from baseline in the monthly attack rate was 89±20.1% (Table S4), and 60% (31 of 52) of the patients were attack-free without the use of long-term prophylaxis. No patients in the lonvo-z group and 2 patients in the placebo group restarted long-term prophylactic therapy.

SAFETY

Adverse events that occurred during or after infusion of lonvo-z or placebo from day 1 through week 28 were reported in 48 of the 52 patients (92%) in the lonvo-z group and in 24 of the 28 patients (86%) in the placebo group (Table 3 and Table S5). The most common adverse event reported in the lonvo-z group was infusion-related reaction, which occurred in 32 patients (62%) in the lonvo-z group and in 5 patients (18%) in the placebo group. All adverse events were grade 1 or 2, no adverse events of grade 3 or higher were reported in either group, and no serious adverse events were reported in the lonvo-z group for the duration of follow-up.

All infusion-related reactions were grade 1 or 2 and were considered by the investigator to be related to lonvo-z or placebo (Table S6); most were transient, occurred shortly after initiation of the infusion, and resolved the same day (Table S7). The most common symptoms associated with infusion-related reactions were headache, flushing, and chest pain or tightness. A total of seven patients in the lonvo-z group had infusion-related reactions that resulted in a decrease in the infusion rate or temporary infusion interruptions. The median time for lonvo-z infusion on day 1 was 3.9 hours (range, 2.5 to 5.3).



No. at Risk														
Lonvo-z	52	52	52	52	52	52	52	50	44	43	20	8	6	3
Placebo	28	28	28	28	27	27	27	25	21	0	0	0	0	0
Placebo to lonvo-z	0	20	20	8	3	0	0	0	0	0	0	0	0	0

Figure 1. Monthly Hereditary Angioedema Attack Rate over Time.

The mean number of hereditary angioedema (HAE) attacks per month before screening was calculated on the basis of the number of patient-reported attacks occurring in the 90 days before the screening visit. The baseline monthly attack rate was defined as the number of HAE attacks per month from the date of the second screening visit to randomization. For patients in the placebo-to-lonvo-z crossover group, the baseline period was defined as the time from the date of the second screening visit to the day before lonvo-z crossover administration or the day before the start of long-term prophylaxis, whichever was earlier. The baseline attack rate and attack rates after infusion were confirmed by the investigator. During the efficacy evaluation period, two patients in the placebo group restarted long-term prophylaxis owing to a high number of attacks. For these patients, data were censored at the time of long-term prophylaxis initiation and only attacks occurring before long-term prophylaxis initiation were included in the analysis. Both patients discontinued long-term prophylaxis before receiving the lonvo-z crossover infusion. I bars indicate the standard error.

Table 3. Adverse Events during or after Infusion of Lonvo-z or Placebo through Week 28.*

Event	Lonvo-z (N=52)	Placebo (N=28)
Any adverse event — no. (%)	48 (92)	24 (86)
Related to trial regimen	37 (71)	6 (21)
Any serious adverse event — no. (%)	0	1 (4)†
Adverse events occurring in >5% of patients in the lonvo-z group — no. (%)		
Infusion-related reaction	32 (62)	5 (18)
Headache	10 (19)	3 (11)
Fatigue	7 (14)	3 (11)
Nasopharyngitis	7 (14)	9 (32)
Back pain	6 (12)	3 (11)
Upper respiratory tract infection	6 (12)	2 (7)
Upper abdominal pain	5 (10)	0
Abdominal pain	4 (8)	0
Influenza	3 (6)	1 (4)
Pain in extremity	3 (6)	0

* All adverse events were graded for severity according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5. No serious adverse events were reported in the lonvo-z group. One event of grade 2, serious supraventricular tachycardia, was reported in the placebo group. Related events refer to those that were deemed by the investigators to be probably or possibly related to lonvo-z or placebo or those for which data regarding the relationship to the trial regimen were missing.

† This patient had serious supraventricular tachycardia, which started on day 39, resolved in 2 days, and was considered by the investigator to be unrelated to the investigational product.

Shifts in the levels of liver enzymes, most of which occurred within 6 weeks after infusion, were transient, asymptomatic, and resolved without intervention (Fig. S6). In the lonvo-z group, elevated levels of aspartate aminotransferase, alanine aminotransferase, or both were observed in eight patients (15%). The levels were all less than 3 times the upper limit of the normal range, with the exception of the levels in one patient (2%), who had an elevated alanine aminotransferase level that fell between 3 times and less than 5 times the upper limit of the normal range with a concurrent upper respiratory tract infection. This patient also had an elevated level of alanine aminotransferase during the screening period. In the placebo group, three patients (11%) had elevations of aspartate aminotransferase or alanine aminotransferase (or both) of less than 3 times the upper limit of the normal range, and no patients had a level that was 3 or more times the upper limit of the normal range. The mean activated partial-thromboplastin time remained similar to baseline levels throughout follow-up (Fig. S7).

PHARMACODYNAMICS, PHARMACOKINETICS, AND IMMUNOGENICITY

Total plasma kallikrein protein levels decreased by the first measurement at week 2 (day 15) in the lonvo-z group, with a mean reduction from baseline of 65±15.7%; in the placebo group, levels increased by 18±19.8% (Fig. 2). Absolute reductions in the total plasma kallikrein protein levels are shown in Figure S8. Steady-state reductions in total plasma kallikrein protein levels were reached by week 5 and were maintained through the last follow-up assessment. The levels of the component analytes of lonvo-z (exogenous lipids [LP000001 and DMG-PEG2k], Cas9 messenger RNA, and a single guide RNA) rapidly declined after infusion, with levels falling below the limit of quantification within 2 to 4 weeks (Fig. S9). After infusion of lonvo-z, transient antidrug antibodies were detected in 8 patients and anti-Cas9 antibodies were detected in 50 patients (Table S8). No discernible effects of these antibodies on pharmacokinetics, pharmacodynamics, or safety were seen.

CROSSOVER ANALYSIS

At the time of the data-cutoff date, 54 patients had opted to receive a second blinded infusion (34 patients in the lonvo-z group received placebo and 20 patients in the placebo group received lonvo-z). The duration of follow-up was limited in patients who received the crossover infusion (median, 1.4 months; range, 0.3 to 4.5). The preliminary results with regard to the efficacy, safety, and pharmacodynamics of lonvo-z among the patients who crossed over from the placebo group to receive a single dose of lonvo-z after week 28 are consistent with those among the patients who had been initially assigned to the lonvo-z group (Fig. 1 and Fig. S10).

DISCUSSION

In this phase 3 trial, a single infusion of 50 mg of lonvo-z resulted in a significantly lower rate of hereditary angioedema attacks than placebo. Patients who received lonvo-z had an 87% lower attack rate than those who received placebo, and 62% of the patients who received lonvo-z were considered by the investigator to be attack-free

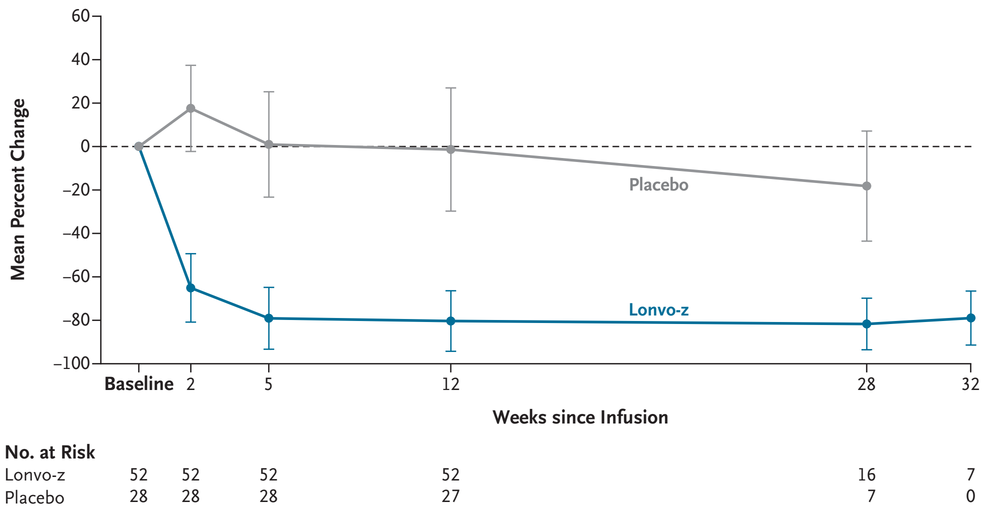


Figure 2. Change from Baseline in Total Plasma Kallikrein Protein Levels.

Shown is the mean percent change from baseline in the total plasma kallikrein protein level in patients receiving either a single infusion of 50 mg of lonvo-z or placebo. In the lonvo-z group, the mean change after infusion was −65±15.7% at week 2 (day 15) (in 52 patients), −79±14.2% at week 5 (in 52 patients), −80±13.9% at week 12 (in 52 patients), and −82±11.9% at week 28 (in 16 patients). I bars indicate standard deviations.

from weeks 5 to 28 without the use of long-term prophylaxis. Treatment with lonvo-z was associated with a reduction in the monthly attack rate, a deep pharmacodynamic response that remained stable over time, and improvements in quality of life. The safety profile of lonvo-z was consistent with that seen in the phase 1–2 trial.[10,13] Adverse events in the lonvo-z group were mild to moderate in severity, with transient infusion-related reactions being the most common adverse event.

The efficacy results were similar to those observed in the phase 1–2 trial. Among patients who received 50 mg of lonvo-z in the open-label phase 1 and blinded phase 2 portions, 3 of 4 patients (75%) and 8 of 11 patients (73%), respectively, were attack-free without the use of long-term prophylaxis at the end of the primary observation period and had mean reductions of greater than 80% in total plasma kallikrein protein levels.[14] Over time, 20 of the 21 patients (95%) who received 50 mg of lonvo-z in the phase 1–2 trial, including 6 crossover patients, became attack-free with no use of long-term prophylaxis.[14] Although the duration of follow-up in the phase 3 trial is limited beyond week 28, with data through 40 weeks for only 8 patients who were assigned to receive lonvo-z, the low attack rate in these patients was maintained.

Hereditary angioedema attacks can be difficult to identify since some symptoms (e.g., abdominal pain) are common and can result from varied causes. This factor presents a challenge in clinical trials that rely on patient-reported symptoms to track the frequency of attacks. The lack of a validated method to definitively attribute such symptoms to the dysregulated kallikrein–kinin system may be a contributing factor as to why some patients do not initially become symptom-free in blinded, placebo-controlled clinical trials of therapies that target this pathway, despite findings of decreased levels of plasma kallikrein protein or activity.[3-5,9]

Nonclinical studies in animal models have suggested that anti-Cas9 immune responses may interfere with the safety or efficacy of gene editing therapies[15-17]; however, the clinical relevance of these findings has not been evaluated. In this trial, as well as in the phase 1 trial of lonvo-z,[10] we found a very low baseline prevalence of anti-Cas9 antibodies with a high incidence of transient anti-Cas9 antibody responses after treatment. Despite this finding, 11 patients who were initially treated with 25 mg of lonvo-z in the phase 1–2 trial were safely treated with a follow-on dose of 50 mg, with evidence of improved clinical responses and deeper reductions in total plasma kallikrein protein levels that were indistinguishable from those observed in patients treated with a single 50-mg dose. This preliminary evidence suggests that the occurrence of anti-Cas9 antibody responses after lonvo-z infusion does not compromise the safety or efficacy of subsequent treatment.

The edit to the gene *KLKB1* is intended to be permanent after treatment with lonvo-z. In a humanized mouse model, persistent editing in hepatocytes through aging and the liver-regeneration process was observed.[10] In the phase 1–2 trial, reduced plasma kallikrein protein levels remained low and stable at the latest follow-up assessment of up to 3 years after one infusion of 50 mg of lonvo-z.[11] These data suggest that the edit in *KLKB1* persists across mitoses (cell division), given the reported rates of hepatocyte turnover.[18]

Consideration must therefore be given to the long-term safety of plasma prekallikrein knockdown. Reports in the literature of a rare congenital plasma kallikrein deficiency (Fletcher factor deficiency) suggest that such persons are healthy and have prolonged activated partial-thromboplastin time without an associated bleeding risk.[19] Efforts have been made to characterize the incidence of certain clinical conditions, such as cardiovascular disorders, among persons with this deficiency.[20,21] However, the rarity of Fletcher factor deficiency, together with reporting bias, makes it difficult to draw definitive conclusions regarding its association with specific clinical outcomes. The data presented here and previously[10,13] indicate that the defining feature of congenital plasma kallikrein deficiency, prolonged activated partial-thromboplastin time, does not develop in patients with hereditary angioedema who are treated with lonvo-z. This may be due to the presence of a low level of residual plasma kallikrein in the context of a C1 inhibitor deficiency. The most prevalent cardiovascular condition reported in persons with congenital plasma kallikrein deficiency is hypertension,[20,22] which has not been reported as an adverse event in patients treated with lonvo-z to date. Consistent with this observation is that no causal association with cardiovascular risk has been reported in patients treated with approved plasma kallikrein–targeted agents.[23,24]

In other clinical trials involving in vivo gene editing, investigational therapies have been associated with acute, sometimes serious or fatal, adverse events, including increased levels of serum alanine aminotransferase and aspartate aminotransferase.[25-30] These investigational agents comprising distinct lipid components and unique active pharmaceutical ingredients are being studied in unique populations of patients, some of whom have serious underlying clinical disease and medically important coexisting conditions, a factor that makes it difficult to generalize safety findings across the therapeutic class. In this trial, no serious adverse events were observed in patients treated with lonvo-z. The most common adverse reactions were infusion-related reactions, which were generally transient and resolved without intervention. Elevated levels of serum aspartate aminotransferase and alanine aminotransferase, which occurred in approximately 10 to 15% of patients treated with lonvo-z, were transient, asymptomatic, and resolved without intervention.

Another important consideration is the potential risk of off-target editing. As gene editing technology continues to advance, each drug candidate needs to be empirically evaluated, first with a comprehensive preclinical assessment, and then by long-term clinical evaluation for the risk of off-target effects. For lonvo-z, extensive computational and laboratory assessments showed that the possibility of clinically relevant off-target editing activity was low.[10] Across all clinical trials conducted with lonvo-z to date, no evidence of adverse effects attributable to off-target editing has been observed. A long-term follow-up study is ongoing to elucidate the long-term safety and efficacy of lonvo-z (NCT06262399).

Limitations of this trial include the limited sample size and length of follow-up as of the data-cutoff date. Owing to these limitations, together with a carefully selected patient sample, it is possible that other risks, particularly in a general population with other coexisting conditions or susceptibilities that may occur outside the controlled setting of a clinical trial, have not yet been identified. Continued follow-up in this trial and its follow-up study will provide more data on long-term outcomes. An additional limitation is the placebo-controlled nature of the trial and the long-term prophylaxis washout requirement, which may have discouraged the participation of some patients. However, the baseline characteristics of the patients in this trial were similar to those of the broader population with hereditary angioedema.[4,5,9,31] Finally, the race and ethnic background of the patients are representative of the countries where the trial was conducted and not of the global population; however, neither race nor ethnicity are known to affect the symptoms of hereditary angioedema.[32]

In this phase 3 trial, among patients with hereditary angioedema due to C1 inhibitor deficiency, a single dose of lonvo-z resulted in a significantly lower rate of hereditary angioedema attacks than placebo from 5 through 28 weeks after infusion. All adverse events were grade 1 or 2 in severity in the lonvo-z group. These data support the use of lonvo-z as a potential treatment for hereditary angioedema.

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SUPPLEMENTARY APPENDIX

Supplement to: Lonvoguran Ziclumeran – in vivo CRISPR Gene Editing in Hereditary Angioedema

Cohn DM, Gurugama P, et al.

This appendix has been provided by the authors to give readers additional information about the work.

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DMC, HJL, EAP, TJC, HF, MM, MAR, DM, AG, MYS, JSB, DL, JL, and AB were involved in study conception and design. DMC, PG, HJL, EAP, TJC, JJ, WRL, MM, JP, MAR, DM, AG, MYS, and AB were involved in data collection. DM, AG, MYS, AS, CRM, AMA, YX, JSB, DL, and JL analyzed the data. DMC, PG, HJL, EAP, TJC, JJ, WRL, MM, JP, MAR, and AB were investigators for the study. The first draft of the manuscript was written by Jeffrey A. Blair, Ph.D., Ellen Woon, Ph.D., and Melissa Austin of Helios Global Group (funded by Intellia Therapeutics). All authors contributed to the data interpretation, review, editing, and approval of the manuscript for publication, and participated in the decision to submit for publication. All authors assume responsibility for the completeness and accuracy of the data and the fidelity of the trial, the manuscript, the protocol (available at NEJM.org), and the statistical analysis plan. All authors had access to all data in the study and agreed to maintain confidentiality of the data during manuscript development.

Supplemental Methods

Crossover

At week 28, all patients had the option to elect to receive a second blinded infusion of the opposite treatment (crossover) from that received on day 1. Patients who did not participate in the crossover followed an alternative schedule of activities at week 28 (see protocol).

Statistical Analysis

For the primary end point, simulations were used to assess the power using a generalized linear model assuming a Poisson distribution with Pearson Chi-square scaling of standard error, which accounted for attack data that are potentially over dispersed. It was assumed that patients in the placebo group would have an average of 3 attacks per month for a total of 18 attacks from weeks 5-28, patients in the lonvo-z group would have an average of 0.75 attacks per month for a total of 4.5 attacks from weeks 5-28, and 60% of patients in the lonvo-z group would be attack-free from week 5 through week 28. In addition, a total of 2.5% and 15% of patients in the lonvo-z arm and the placebo arm, respectively, would discontinue the study prior to week 28 assessment. Based on these assumptions, a total of 60 patients (40 in the lonvo-z arm and 20 in the placebo arm) would provide 99% power to reject the null hypothesis at a 2-sided alpha of 0.05 when the true hereditary angioedema (HAE) attack rate ratio is 0.25 (lonvo-z vs placebo).

Formal statistical hypothesis testing was performed on the primary and key secondary efficacy end points with the global family-wise type I error rate strongly controlled at a 2-sided 0.05 alpha level using a hierarchical testing procedure. Specifically, the hypothesis on the primary end point was first tested at a 2-sided alpha of 0.05. The hypotheses on the key secondary end points were formally tested at the same alpha level in the following order if the primary end point was found to be statistically significant:

- Time-normalized number of investigator-confirmed HAE attacks requiring on-demand therapy from week 5 through week 28
- Time-normalized number of moderate or severe investigator-confirmed HAE attacks from week 5 through week 28
- Investigator-confirmed HAE attack-free status from week 5 through week 28
- Change from baseline to week 28 in Angioedema Quality of Life (AE-QoL) Questionnaire total score.

All other secondary and exploratory end points were evaluated in a descriptive manner.

For HAE attack-related end points, baseline was defined as the time from the screening 2 visit to the date of randomization. For AE-QoL end point, baseline was defined as the last assessment prior to randomization. For safety and pharmacokinetic end points, baseline was defined as the last assessment prior to the administration of study intervention. For plasma kallikrein protein levels, the average value of 3 samples collected prior to study intervention administration was used as baseline.

For patients who initiated long-term prophylaxis prior to the week 28 assessment, only attacks with onset and AE-QoL data collected prior to the initiation of long-term prophylaxis were included in the analysis (hypothetical strategy). All data were included in the analysis regardless of the use of any on-demand medication or short-term prophylaxis (treatment policy strategy).

Of the 16 patients who had not yet reached week 28 at the time of the data cutoff, complete 28-week outcome data were unavailable. To evaluate the robustness of the primary and key secondary endpoint analyses, the following prespecified supplementary and sensitivity analyses were conducted per the statistical analysis plan:

- A supplementary analysis was performed using the Efficacy Analysis Set, defined as all randomized patients who received at least a partial dose of study intervention and had week 28 assessment.
- Sensitivity analyses using multiple imputation (MI) were performed under both a missing-at-random (MAR) assumption and a more conservative jump-to-reference (J2R) assumption.

In addition, a supplementary analysis was performed for the primary endpoint using the Per-Protocol Analysis Set.

Results from all supplementary and sensitivity analyses were highly consistent with those of the main analyses, supporting the robustness of the findings, and are not reported here because they are largely redundant with what is already presented and do not influence the interpretation of the efficacy outcome.

Plasma concentrations and concentration-time profiles were tabulated for the components of lonvo-z (exogenous lipids [DMG-PEG2k and LP000001], Cas9 mRNA, and *KLKB1* single guide RNA [sgRNA]). Antidrug antibodies to lonvo-z and anti-Cas9 were summarized descriptively.

Supplemental Results

Treatment Crossover

At the time of the data cutoff date, 54 patients had opted to receive a second blinded infusion (lonvo-z to placebo, n=34; placebo to lonvo-z, n=20).

One patient (placebo to lonvo-z) permanently discontinued the infusion and did not receive the full dose (2%, 1 mg lonvo-z infused) due to a grade 2 infusion-related reaction (IRR), which began shortly after the start of infusion and included symptoms of lightheadedness, headache, and flushing. The infusion was paused and symptoms resolved within 13 minutes of onset without treatment. The infusion was restarted at a reduced infusion rate but then permanently discontinued shortly thereafter per investigator decision. The symptoms and severity of this event were consistent with other reported IRRs (Longhurst HJ, et al. *N Engl J Med* 2024;390:432-44; Cohn DM, et al. *N Engl J Med* 2025;392:458-67).

Two patients (both placebo to lonvo-z) did not receive the full lonvo-z dose at crossover due to reaching the infusion expiry time. One patient experienced a grade 2 IRR; symptoms included shortness of breath (dyspnea), tachycardia, chest pain, and facial swelling (lips). The infusion was interrupted to permit symptoms to resolve, and the patient received an additional dose of dexamethasone to manage symptoms. The infusion resumed at a reduced infusion rate, which was well tolerated by the patient, but ultimately could not be completed due to reaching the infusion expiry time (31%, 15.5 mg lonvo-z infused). The other patient had multiple infusion interruptions due to line occlusions, which contributed to reaching the infusion expiry time (79%, 39.5 mg lonvo-z infused).

Figure S1. HAELO study design

HAELO is a phase 3, placebo-controlled, double-blind, randomized study. At the screening 1 visit, patients signed the informed consent form and started LTP washout. Patients must washout prior LTP from within 5 half-lives prior to the start of the run-in period. Once washout was completed, patients had a screening 2 visit when they entered the run-in period. Access to on-demand treatment was required at all times during the study, including the screening/run-in. Patients were randomized 2:1 and stratified by the baseline number of investigator-confirmed hereditary angioedema attacks per month (>3 vs ≤3, month defined as 28 days) determined during the run-in period. Infusion of lonvo-z or placebo occurred on day 1 and was administered intravenously over approximately 4 hours. At week 28, patients may have elected to receive a blinded infusion of the opposite treatment from that received at randomization (crossover). Patients are followed through week 104 at which point all patients are invited to enroll in the long-term follow-up study (NCT06262399). LTP, long-term prophylaxis.

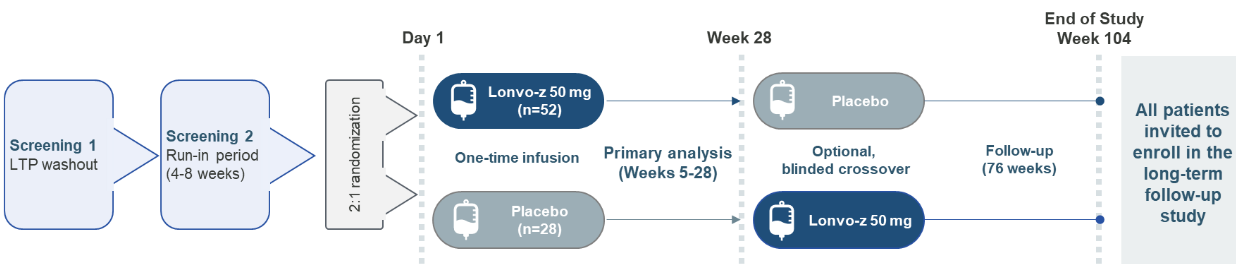


Figure S2. CONSORT diagram
Flowchart of screening, enrollment, and allocation of patients in the phase 3 study of lonvo-z vs placebo in patients with hereditary angioedema. Follow-up of all 80 patients in the study is ongoing as of the data cutoff date, February 10, 2026. One patient in the placebo arm completed the first week 28 visit on the day of the data cutoff date but did not receive the crossover infusion until the day after the data cutoff date.

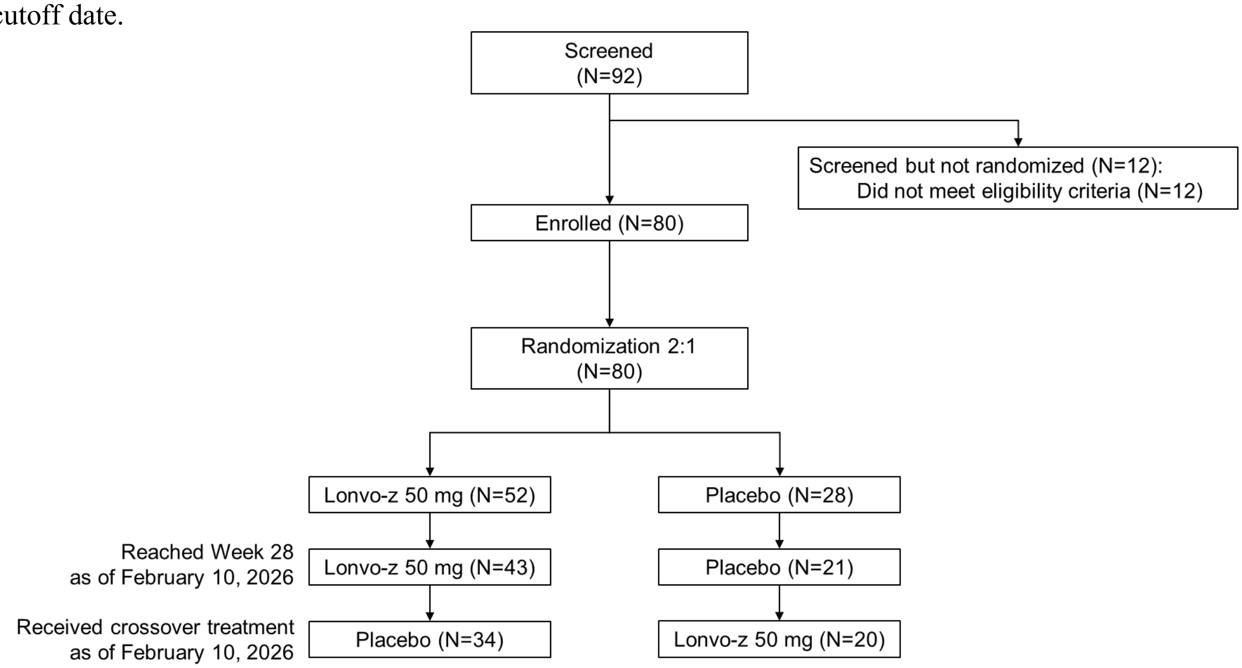


Figure S3. Hereditary angioedema attacks over time

Horizontal blue bars represent the interval between the beginning of the run-in period for the individual patient and the last assessment of that patient as of the data cutoff date. The severity and length of angioedema attacks in individual patients during the 4- to 8-week screening period, the 28-week primary observation period, and the postprimary observation period are represented by colored bars, with the length of each bar indicating the duration of the attack. Patient data are presented from top to bottom according to the study group, attack-free status, and length of follow-up. Of the 38% (20/52) of patients in the lonvo-z arm who reported attacks from week 5-28, patients had a mean attack rate reduction of 72% (SD 24.6). In patients who received placebo who continued to experience attacks (25/28), patients had a mean 21% (SD 40.3) reduction of attacks. Two patients in the placebo group took concomitant LTP therapy due to a high rate of attacks during the primary observation period. Angioedema attacks occurred in multiple anatomical locations (abdominal, peripheral, laryngeal) in both groups. At week 28, all patients had the option to elect to receive a second blinded infusion of the opposite treatment (crossover) of that received on day 1. LTP, long-term prophylaxis; pt, patient.

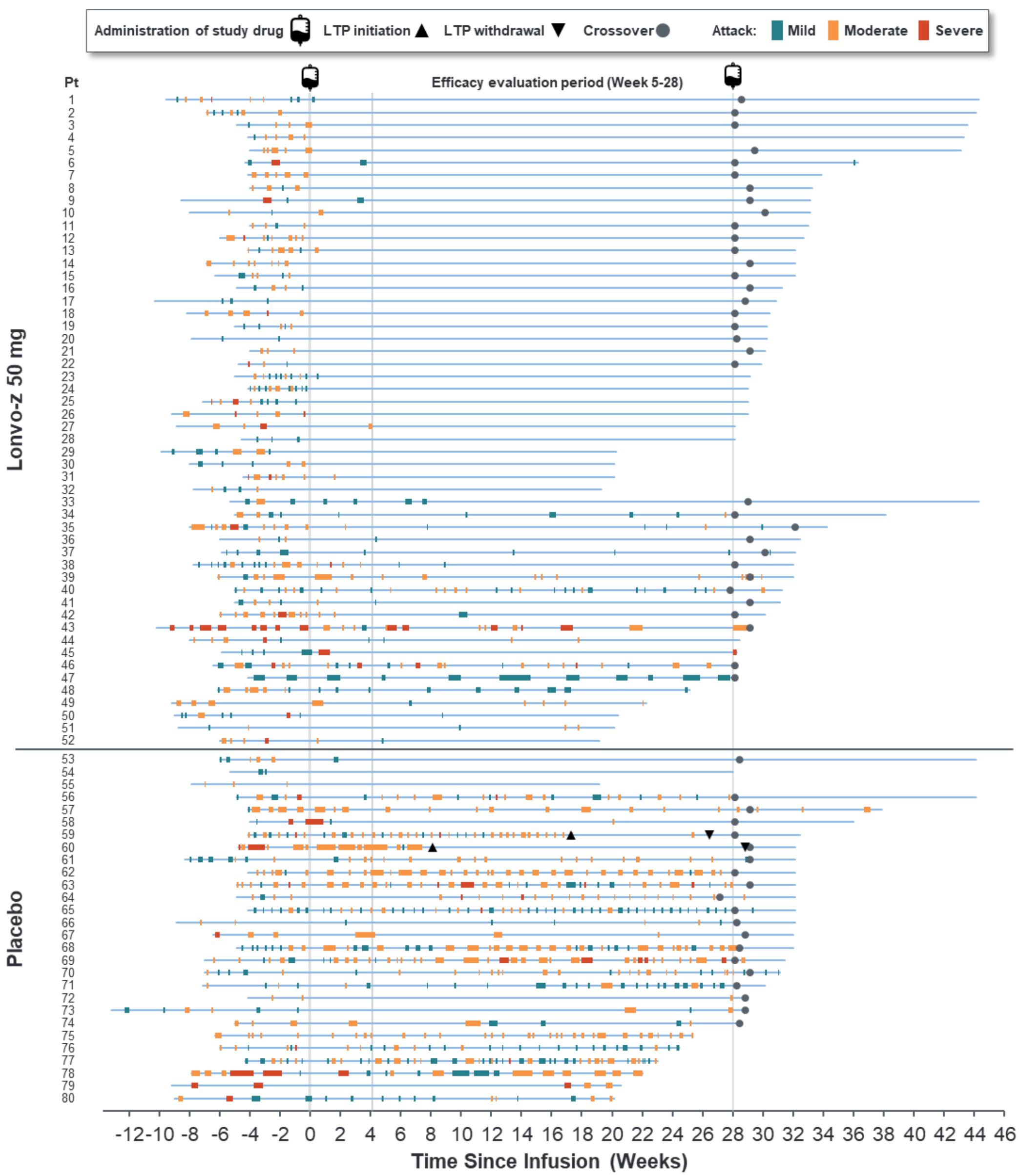
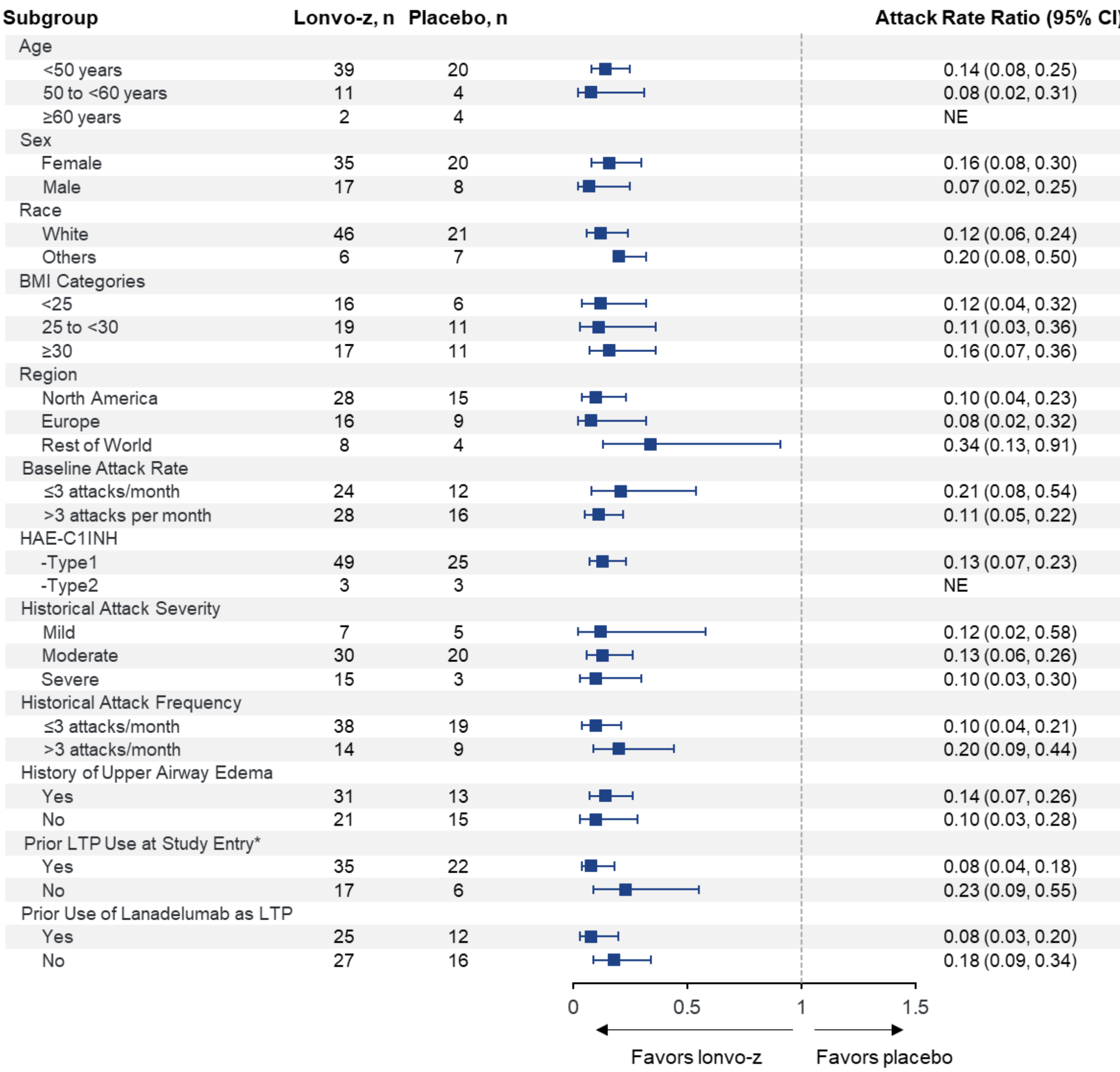


Figure S4. Attack rate reduction subgroup analyses (week 5 through week 28)

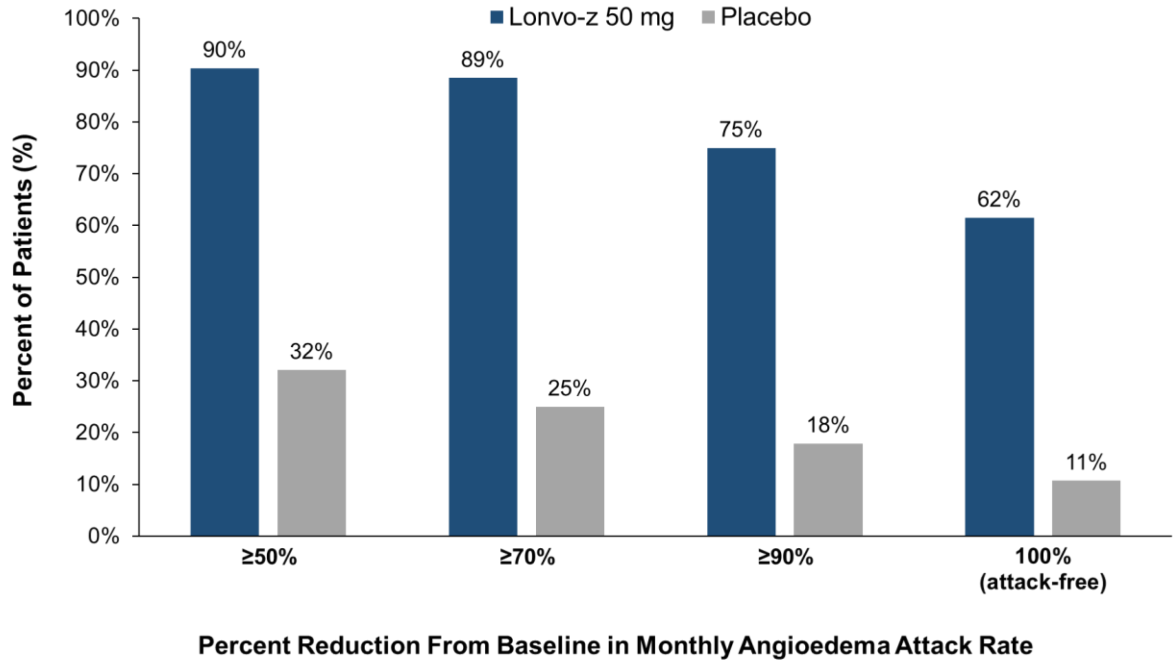


Subgroup	Lonvo-z, n	Placebo, n	Attack Rate Ratio (95% CI)
Age			
<50 years	39	20	0.14 (0.08, 0.25)
50 to <60 years	11	4	0.08 (0.02, 0.31)
≥60 years	2	4	NE
Sex			
Female	35	20	0.16 (0.08, 0.30)
Male	17	8	0.07 (0.02, 0.25)
Race			
White	46	21	0.12 (0.06, 0.24)
Others	6	7	0.20 (0.08, 0.50)
BMI Categories			
<25	16	6	0.12 (0.04, 0.32)
25 to <30	19	11	0.11 (0.03, 0.36)
≥30	17	11	0.16 (0.07, 0.36)
Region			
North America	28	15	0.10 (0.04, 0.23)
Europe	16	9	0.08 (0.02, 0.32)
Rest of World	8	4	0.34 (0.13, 0.91)
Baseline Attack Rate			
≤3 attacks/month	24	12	0.21 (0.08, 0.54)
>3 attacks per month	28	16	0.11 (0.05, 0.22)
HAE-C1INH			
-Type1	49	25	0.13 (0.07, 0.23)
-Type2	3	3	NE
Historical Attack Severity			
Mild	7	5	0.12 (0.02, 0.58)
Moderate	30	20	0.13 (0.06, 0.26)
Severe	15	3	0.10 (0.03, 0.30)
Historical Attack Frequency			
≤3 attacks/month	38	19	0.10 (0.04, 0.21)
>3 attacks/month	14	9	0.20 (0.09, 0.44)
History of Upper Airway Edema			
Yes	31	13	0.14 (0.07, 0.26)
No	21	15	0.10 (0.03, 0.28)
Prior LTP Use at Study Entry*			
Yes	35	22	0.08 (0.04, 0.18)
No	17	6	0.23 (0.09, 0.55)
Prior Use of Lanadelumab as LTP			
Yes	25	12	0.08 (0.03, 0.20)
No	27	16	0.18 (0.09, 0.34)

*Prior use of LTP at study entry was defined as the last dose of LTP being discontinued within 100 days of screening 1 visit.
Attack rate ratio was not estimable for age group ≥60 years and HAE-C1INH Type 2 due to small sample sizes. All 3 patients ≥60 years in the lonvo-z group were attack-free from day 1 infusion through latest follow-up and all 3 patients with HAE-C1INH Type 2 in the lonvo-z group were attack-free from day 1 infusion through latest follow-up.
BMI, body mass index; HAE-C1INH, hereditary angioedema due to C1 inhibitor deficiency; LTP, long-term prophylaxis; NE, not estimable.

Figure S5. Hereditary angioedema attack rate reduction

A. Week 5 through week 28



B. Week 1 through week 28

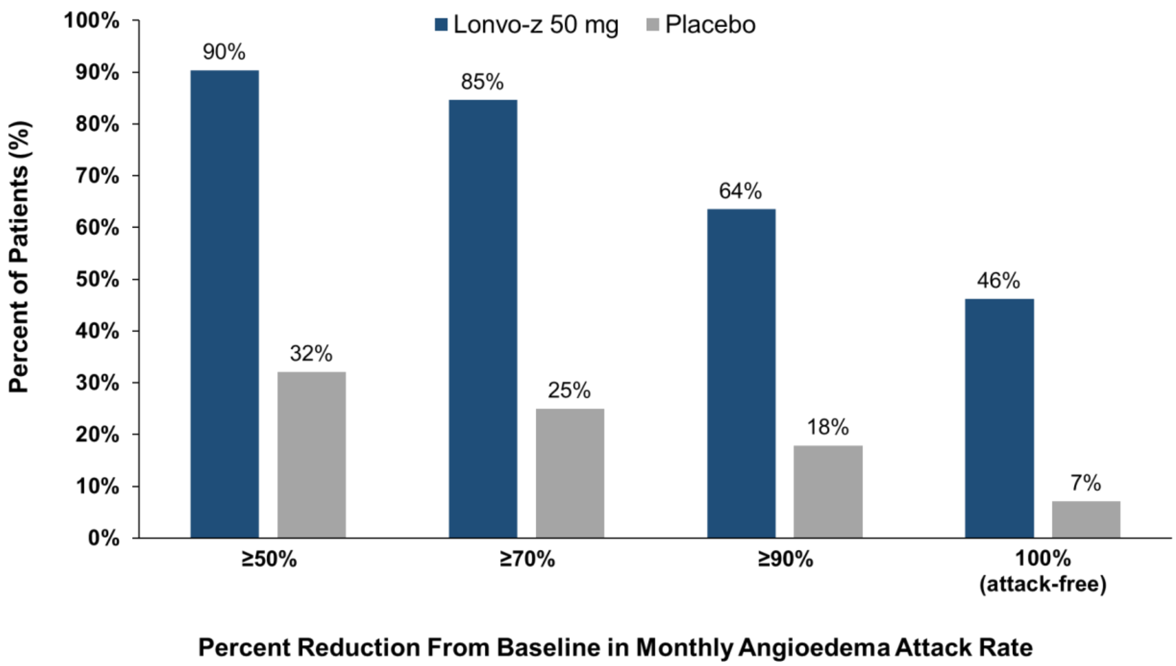
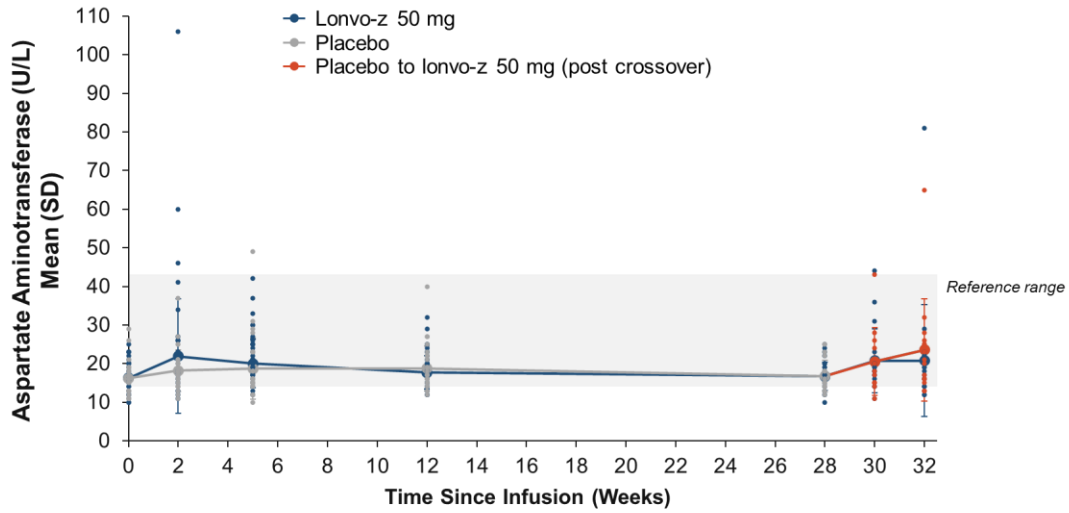


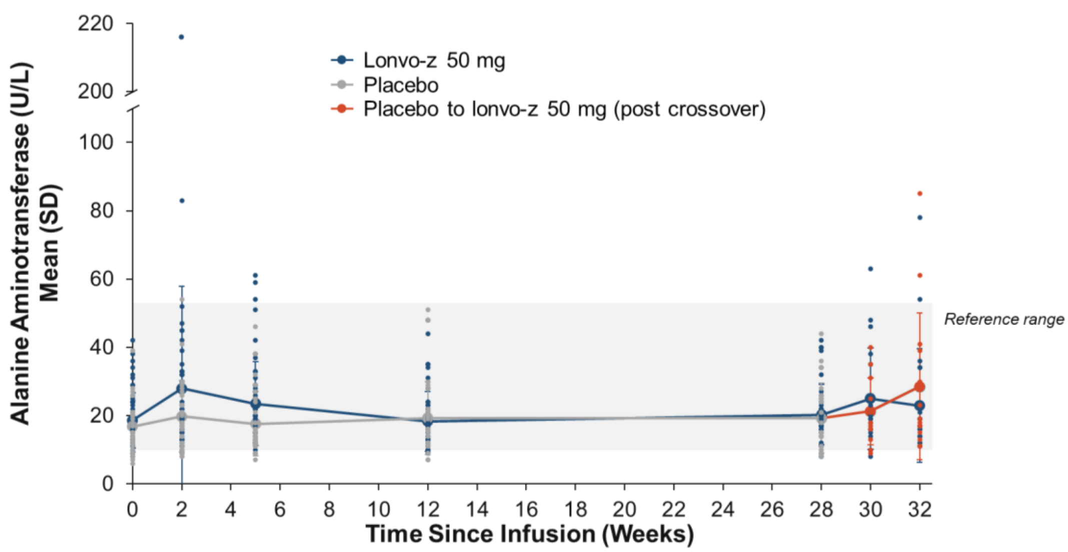
Figure S6. Aspartate aminotransferase and alanine aminotransferase over time
Absolute aspartate aminotransferase and alanine aminotransferase levels at baseline and at scheduled study visits after infusion.

A. Absolute aspartate aminotransferase levels



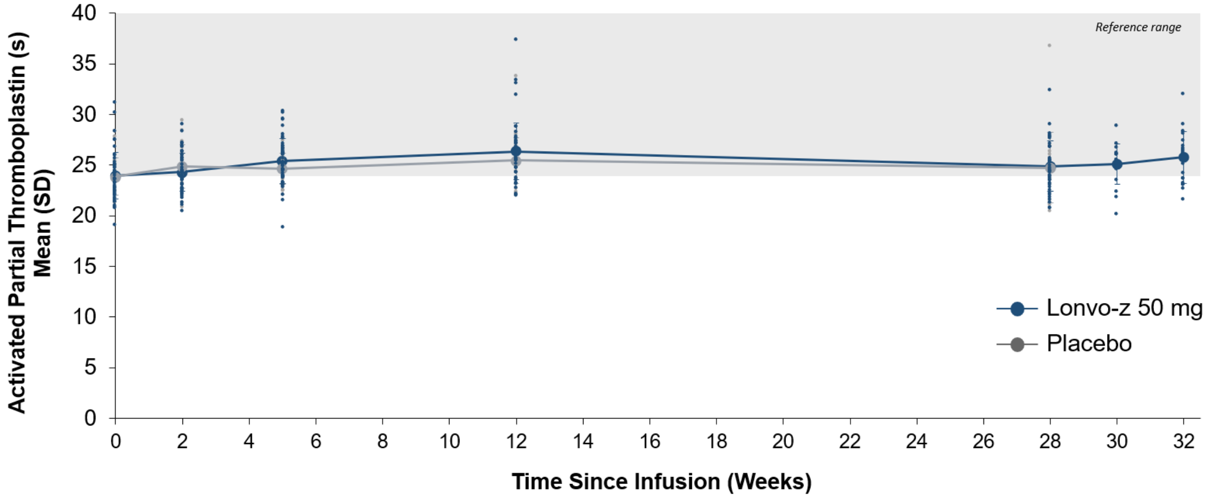
	0	2	5	12	28	30	32
Lonvo-z 50 mg (n)	52	52	52	52	40	18	21
Placebo (n)	28	28	28	28	19	0	0
Placebo to lonvo-z 50 mg (n)	0	0	0	0	0	12	14

B. Absolute alanine aminotransferase levels



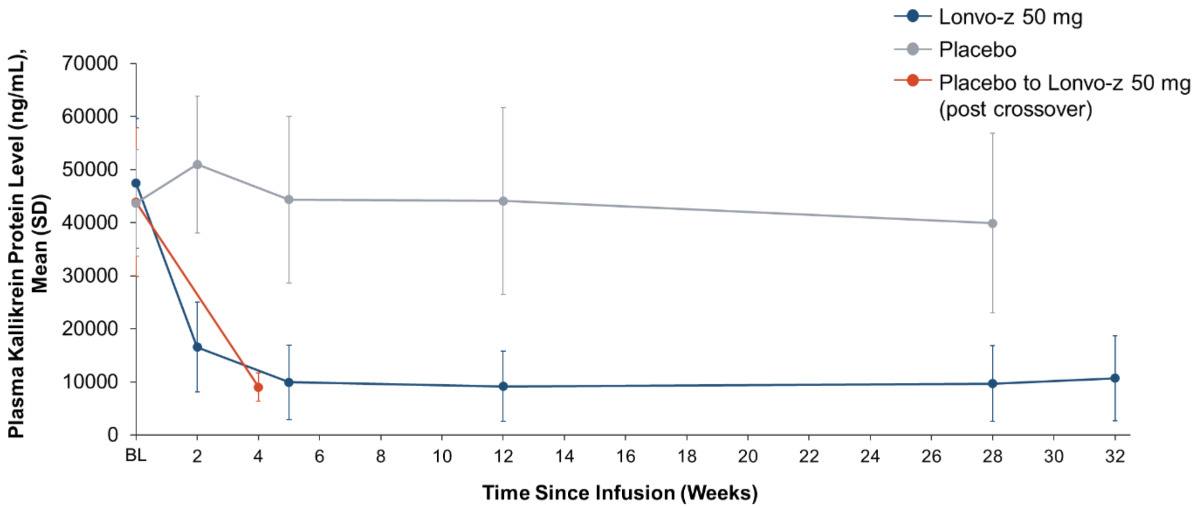
	0	2	5	12	28	30	32
Lonvo-z 50 mg (n)	52	52	52	52	40	18	21
Placebo (n)	28	28	28	28	19	0	0
Placebo to lonvo-z 50 mg (n)	0	0	0	0	0	12	14

Figure S7. Activated partial thromboplastin time over time



Lonvo-z 50 mg (n)	52	50	51	52	40	19	21
Placebo (n)	28	28	28	28	20	0	0

Figure S8. Absolute reductions in total plasma kallikrein protein in patients in the lonvo-z group, placebo group, and placebo to lonvo-z group



	BL	2	4	5	12	28	32
Lonvo-z 50 mg (n)	52	52		52	52	16	7
Placebo (n)	28	28		28	27	7	0
Placebo to Lonvo-z 50 mg (n)	20		3		0	0	0

Figure S9. Plasma concentrations of lonvo-z analytes following 50 mg infusion

Analytes rapidly declined from peak plasma concentrations at the end of lonvo-z 50 mg infusion, followed by log-linear elimination to below limit of quantification within 2-4 weeks.

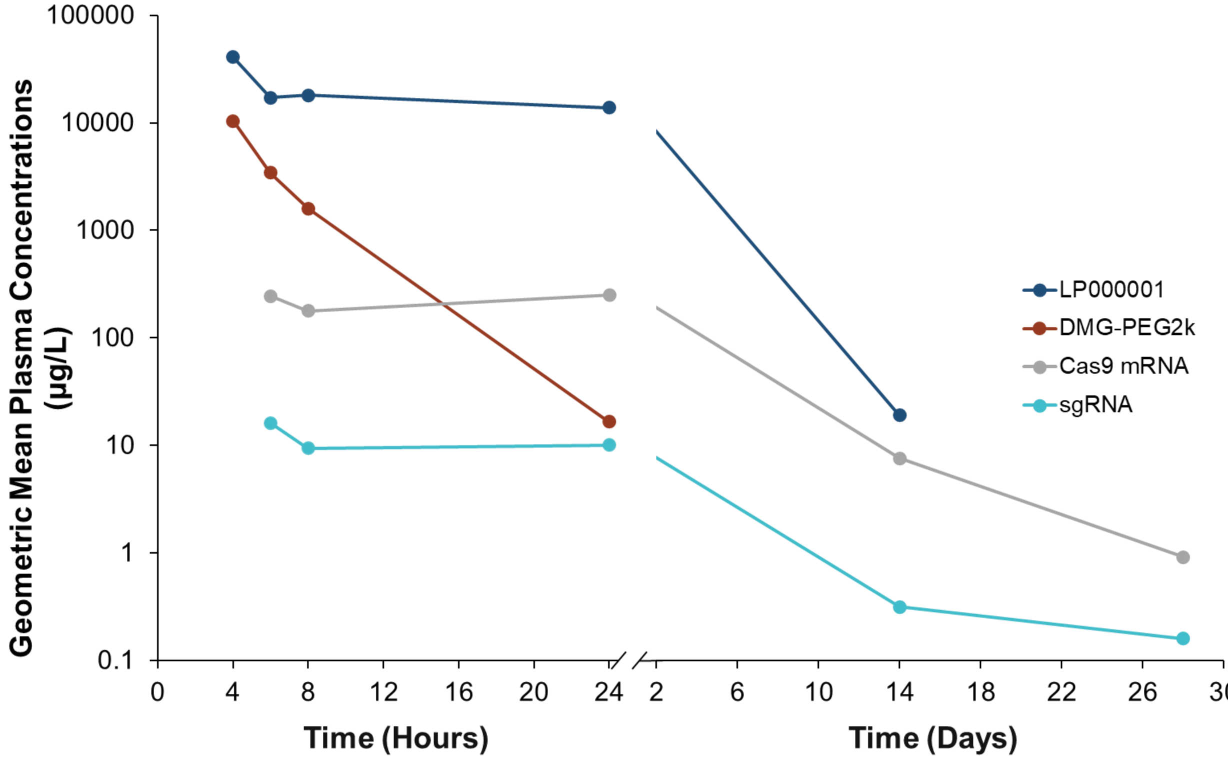
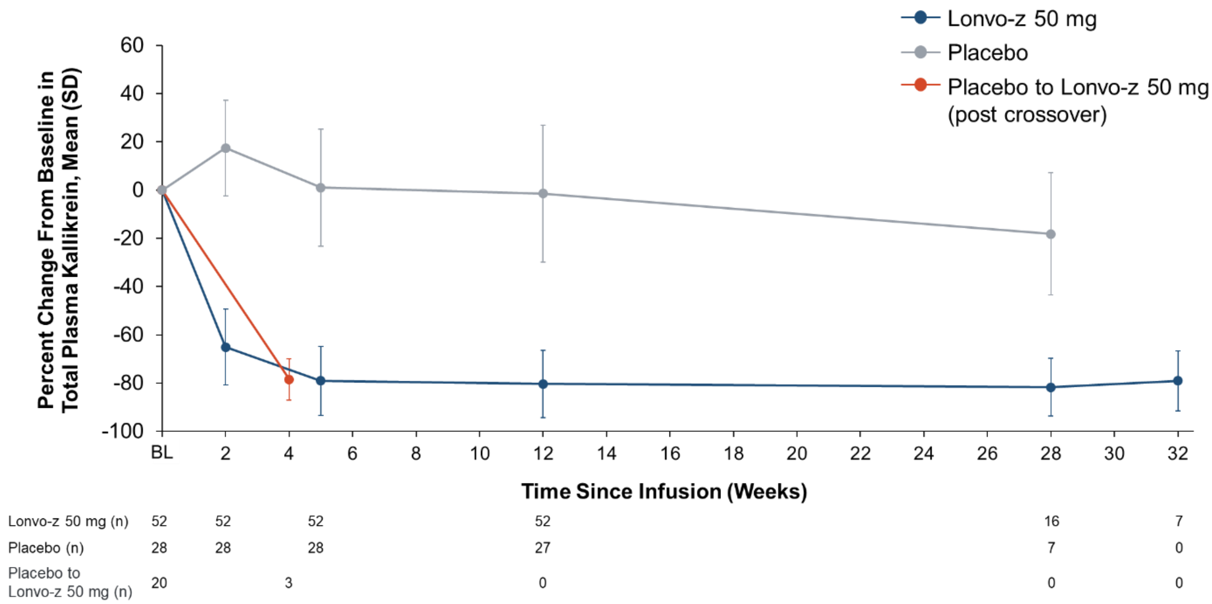


Figure S10. Change from baseline in total plasma kallikrein protein level in patients in the lonvo-z group, placebo group, and placebo to lonvo-z group



Baseline is the average values of samples collected prior to the administration of study intervention. For patients in the placebo group who received lonvo-z as crossover, baseline is the average of 3 most recent samples collected prior to lonvo-z crossover administration. BL, baseline.

Table S1. Patient demographics and baseline characteristics*

Characteristic	Lonvo-z 50 mg (N=52)	Placebo (N=28)	All Patients (N=80)
Median age at screening (range) — yr	42.0 (23.0–71.0)	40.0 (19.0–76.0)	41.5 (19.0–76.0)
Sex — no. (%)			
Female	35 (67)	20 (71)	55 (69)
Male	17 (33)	8 (29)	25 (31)
Ethnicity — no. (%)			
Hispanic or Latino	4 (8)	1 (4)	5 (6)
Not Hispanic or Latino	47 (90)	27 (96)	74 (93)
Unknown	1 (2)	0 (0)	1 (1)
Race — no. (%)†			
White or Caucasian	46 (89)	21 (75)	67 (84)
Black or African American	1 (2)	1 (4)	2 (3)
American Indian or Alaska native	1 (2)	0 (0)	1 (1)
Multiple	3 (6)	3 (11)	6 (8)
Not reported	1 (2)	2 (7)	3 (4)
Unknown	0 (0)	1 (4)	1 (1)
HAE-C1INH — no. (%)‡			
Type 1	49 (94)	25 (89)	74 (93)
Type 2	3 (6)	3 (11)	6 (8)
HAE treatment prior to study enrollment — no. (%)§			
LTP	35 (67)	22 (79)	57 (71)
ODT only	17 (33)	6 (21)	23 (29)
LTP use prior to study enrollment — no. (%)§			
Lanadelumab	25 (48)	12 (43)	37 (46)
C1 inhibitor	5 (10)	3 (11)	8 (10)
Berotralstat	4 (8)	1 (4)	5 (6)
Garadacimab	1 (2)	3 (11)	4 (5)
Other‖	2 (4)	3 (11)	5 (6)
Best response to prior LTP — no. (%)			
Complete control (no attacks)	7 (14)	9 (32)	16 (20)
Partial control	25 (48)	9 (32)	34 (43)
No response	1 (2)	3 (11)	4 (5)
Unknown	2 (4)	1 (4)	3 (4)
Typical attack severity — no. (%)			
Mild	7 (14)	5 (18)	12 (15)
Moderate	30 (58)	20 (71)	50 (63)
Severe	15 (29)	3 (11)	18 (23)
Mean number of HAE attacks per month in the 3 months prior to screening (SD)	1.8 (2.07)	1.5 (1.73)	1.7 (1.95)
Mean number of HAE attacks per month during the run-in period (SD)	3.51 (1.79)	3.54 (1.91)	3.52 (1.82)
Mean AE-QoL total score at baseline (SD)	58.4 (18.05)	58.0 (12.54)	58.3 (16.25)

*Percentages may not total 100 because of rounding.
†Race was self-reported by the patient.

‡Hereditary angioedema with low C1 inhibitor levels (HAE-C1INH-Type 1) or impaired C1 inhibitor function (HAE-C1INH-Type 2).
§LTP prior to study enrollment was defined as the last dose of LTP being discontinued within 100 days of the Screening-1 visit.
‖Other LTP included 'drugs used in hereditary angioedema,' tranexamic acid, and danazol.
AE-QoL, Angioedema-Quality of Life; C1INH, C1 inhibitor; HAE, hereditary angioedema; LTP, long-term prophylaxis; ODT, on-demand treatment; SD, standard deviation.

Table S2. Representativeness of study patients

Category	
Disease under investigation	Hereditary angioedema due to C1 inhibitor deficiency (HAE-C1INH)
Special considerations related to:	
Sex and gender	Males and females are affected equally at the genetic level. Symptoms are sometimes more severe in female patients, possibly due to fluctuation in female sex hormones, and this may contribute to a slightly higher likelihood of diagnosis.
Age	HAE is an autosomal-dominant genetic disease where patients have the disease from birth. The age of onset of attacks is variable, but the disease commonly presents in childhood or adolescence. Patients continue having attacks throughout their lifetime.
Race or ethnic group	There are no known differences in prevalence, disease pathophysiology, or responses to treatment among racial or ethnic groups.
Geography	There are no known differences in prevalence in different countries. HAE is a rare disease (1:50,000 population) and shares symptoms with other more common diseases; therefore, this may contribute to patients being misdiagnosed or underdiagnosed in countries where access to diagnostic tests and evidence-based therapies is difficult or lacking.
Other considerations	HAE-C1INH is divided into 2 types: HAE-C1INH Type 1 (≈85% of patients) and HAE-C1INH Type 2 (≈15% of patients). Attacks most often affect the skin, gastrointestinal tract, and upper airway. The frequency and severity of attacks is highly variable across patients.
Overall representativeness of this study	This was a phase 3 study of 80 adult patients with HAE-C1INH. The age distribution of study patients was consistent with what is observed in adults with HAE-C1INH. There was a lower number of men than women. Patients were representative of patient populations in countries with study sites, which have predominantly White or Caucasian populations. The number of patients with HAE-C1INH Type 1 and HAE-C1INH Type 2 disease was representative of the overall HAE-C1INH population. Patients had varying frequency and severity of HAE attacks, which is typical in patients with HAE.

HAE, hereditary angioedema.

Table S3. Investigator-confirmed HAE attacks week 1 through week 28

Endpoint (week 1-week 28)	Lonvo-z 50 mg (n=52)	Placebo (n=28)
Number of investigator-confirmed HAE attacks per month—LS mean (95% CI)*,†	0.32 (0.20, 0.51)	2.07 (1.55, 2.75)
Percent difference vs placebo (95% CI)	-84% (-91%, -75%)	
Number of investigator-confirmed HAE attacks requiring on-demand therapy— LS mean (95% CI)*,†	0.23 (0.14, 0.40)	1.73 (1.25, 2.39)
Percent difference vs placebo (95% CI)	-86% (-92%, -76%)	
Number of moderate or severe HAE attacks—LS mean (95% CI)*,†	0.14 (0.08, 0.26)	1.20 (0.84, 1.70)
Percent difference vs placebo (95% CI)	-88% (-93%, -78%)	

*The mean number of HAE attacks per month is estimated using a Poisson regression model with Pearson chi-square scaling of standard error with treatment arm and baseline attack rate (>3 vs ≤3 attacks per month) as covariates. Baseline is defined as the time from date of screening 2 visit to randomization. A month is defined as 28 days.

†For patients who initiated any long-term prophylaxis before the week 28 assessment, only attacks that started prior to prophylaxis initiation were included in the analysis. For patients who did not reach week 28 assessment, all data through their latest attack assessment were included.

CI, confidence interval; HAE, hereditary angioedema; LS, least squares.

Table S4. Descriptive summary of monthly rate of investigator-confirmed attacks

	Lonvo-z 50 mg (n=52)	Placebo (n=28)
Monthly HAE attack rate, mean (SD)		
Baseline	3.51 (1.79)	3.54 (1.91)
Week 5 through 28	0.33 (0.64)	2.70 (2.24)
Percent change from baseline	-89% (20.29)	-29% (45.39)
Week 5 through latest assessment	0.32 (0.63)	2.72 (2.23)
Percent change from baseline	-89% (20.06)	-29% (45.73)
Monthly HAE attack rate requiring on-demand therapy, mean (SD)		
Baseline	2.82 (2.10)	2.81 (2.11)
Week 5 through 28	0.25 (0.57)	2.41 (2.29)
Percent change from baseline	-89% (27.84)	-6% (90.16)
Week 5 through latest assessment	0.24 (0.56)	2.43 (2.28)
Percent change from baseline	-90% (27.80)	-4% (92.01)
Monthly moderate or severe HAE attack rate, mean (SD)		
Baseline	2.21 (1.53)	2.23 (1.39)
Week 5 through 28	0.16 (0.43)	1.78 (1.63)
Percent change from baseline	-92% (22.63)	-15% (69.88)
Week 5 through latest assessment	0.16 (0.44)	1.78 (1.62)
Percent change from baseline	-91% (22.89)	-15% (68.08)

Baseline is defined as the time from date of screening 2 visit to randomization. A month is defined as 28 days.

The descriptive summary of HAE attacks per month was calculated for each patient as the number of HAE attacks occurring during Week 5 to Week 28 divided by number of days the patient contributed to the period and multiplied by 28 days.

For patients who initiated any long-term prophylaxis after receiving study intervention, data were censored at the time of long-term prophylaxis initiation, and only attacks that started prior to long-term prophylaxis initiation were included in the analysis.

HAE, hereditary angioedema; SD, standard deviation.

Table S5. Summary of treatment-emergent adverse events within 1 day, within 28 days, and >28 days post infusion

Reported in >5% of patients in the lonvo-z group	Lonvo-z 50 mg (n=52)	Placebo to Lonvo-z (n=20)	Placebo (n=28)
TEAEs within 1 day post infusion — no. (%)			
Infusion-related reaction	32 (62)	9 (45)	5 (18)
Fatigue	5 (10)	1 (5)	1 (4)
Headache	4 (8)	1 (5)	1 (4)
TEAEs within 28 days post infusion — no. (%)			
Infusion-related reaction	32 (62)	9 (45)	5 (18)
Headache	7 (14)	2 (10)	2 (7)
Fatigue	6 (12)	1 (5)	1 (4)
Back pain	4 (8)	2 (10)	2 (7)
Abdominal pain	3 (6)	0	0
TEAEs more than 28 days post infusion — no. (%)*			
Nasopharyngitis	7 (14)	0	9 (32)
Upper respiratory tract infection	6 (12)	1 (5)	2 (7)
Headache	5 (10)	0	1 (4)
Abdominal pain upper	4 (8)	0	0
Influenza	4 (8)	0	1 (4)
Fatigue	3 (6)	0	2 (7)
Gastroenteritis	3 (6)	0	2 (7)

*TEAEs that occurred more than 28 days post infusion are defined as those with a start date later than the date of study intervention administration plus 28 days. For patients who received placebo on day 1 and lonvo-z as crossover, the TEAE start date is relative to the date of lonvo-z crossover administration.
TEAE, treatment-emergent adverse event.

Table S6. Summary of treatment-related treatment-emergent adverse events

Treatment-related-emergent adverse events reported in >5% of patients in the lonvo-z group	Lonvo-z 50 mg (n=52)	Placebo to Lonvo-z (n=20)	Placebo (n=28)
Any treatment-related adverse event	39 (75)	10 (50)	6 (21)
Infusion-related reaction	32 (62)	9 (45)	5 (18)
Fatigue	5 (10)	1 (5)	2 (7)
Headache	3 (6)	0	0

Table S7. Summary of infusion-related reactions

	Lonvo-z (n=52)	Placebo – lonvo-z (n=20)	Lonvo-z Total (n=72)	Placebo (n=28)
Any infusion-related reaction — no. (%)	32 (62)	9 (45)	41 (57)	5 (18)
Median time to onset (range), h	0.2 (0.0–7.6)	0.2 (0.0–4.7)	0.2 (0.0–7.6)	1.0 (1.0–7.6)
Median duration (range), h	11.1 (0.1–174.0)	1.3 (0.0–33.7)	5.3 (0.0–174.0)	18.0 (6.6–54.1)
Resolved ≤24 h since onset — no. (%)*	23 (72)	8 (89)	31 (76)	4 (80)
Resolved >24 h since onset — no. (%)*	9 (28)	1 (11)	10 (24)	1 (20)
Symptoms in ≥5% patients in lonvo-z total group — no. (%)				
Headache	14 (27)	2 (10)	16 (22)	3 (11)
Flushing	9 (17)	5 (25)	14 (19)	1 (4)
Chest pain/tightness	7 (14)	4 (20)	11 (15)	1 (4)
Fever	8 (15)	1 (5)	9 (13)	0
Nausea	8 (15)	1 (5)	9 (13)	0
Back pain	6 (12)	2 (10)	8 (11)	2 (7)
Myalgia	6 (12)	0	6 (8)	0
Fatigue	5 (10)	1 (5)	6 (8)	2 (7)
Chills/shivering	4 (8)	2 (10)	6 (8)	0
Tachycardia	3 (6)	2 (10)	5 (7)	0
Shortness of breath (dyspnea)	3 (6)	1 (5)	4 (6)	1 (4)
Lightheadedness	2 (4)	2 (10)	4 (6)	2 (7)
Infusion-related reaction management — no. (%)				
No change to infusion	19 (37)	4 (20)	23 (32)	5 (18)
At least one dose modification	7 (14)	4 (20)	11 (15)	0
Infusion interruption	6 (12)	3 (15)	9 (13)	0
Infusion rate reduced	1 (2)	0	1 (1)	0
Infusion discontinuation	0	1 (5)	1 (1)	0
Infusion modification not applicable (IRR after infusion)	6 (12)	1 (5)	7 (10)	0
Use of concomitant medications	21 (40)	5 (25)	26 (36)	2 (7)
Acetaminophen	14 (27)	3 (15)	17 (24)	2 (7)
Ibuprofen	8 (15)	0	8 (11)	0
Ondansetron	3 (6)	0	3 (4)	0
Cetirizine	2 (4)	1 (5)	3 (4)	1 (4)
Acetylsalicylic acid	0	1 (5)	1 (1)	0
Alprazolam	1 (2)	0	1 (1)	0
Dexamethasone phosphate	0	1 (5)	1 (1)	0

Diphenhydramine	1 (2)	0	1 (1)	0
Famotidine	1 (2)	0	1 (1)	0
Hydrocortisone	1 (2)	0	1 (1)	0
Mometasone furoate; olopatadine hydrochloride	1 (2)	0	1 (1)	0
Naproxen sodium	0	1 (5)	1 (1)	0
Prednisone	1 (2)	0	1 (1)	0
Sodium chloride	1 (2)	0	1 (1)	0
Bilastine	0	0	0	1 (4)
Rupatadine fumarate	0	0	0	1 (4)

*The denominator is the number of patients with at least 1 IRR. Resolution categories for patients with multiple IRRs were determined by the longest duration event. IRR symptoms occurring in more than 5% of patients in the lonvo-z group and lasting >24 hours to 48 hours were headache, fever, nausea, back pain, myalgia, and tachycardia. Symptoms of flushing in two patients and fatigue in one patient lasted more than 48 hours. The maximum reported duration was 100 hours (4.2 days) for flushing and 174 hours (7.3 days) for fatigue.
IRR, infusion-related reaction.

Table S8. Serum antidrug antibodies and anti-Cas9 protein antibodies

Antidrug antibodies represent IgG and IgM antibodies to the lonvo-z lipid nanoparticle. Anti-Cas9 antibodies represent IgG and IgM antibodies to Cas9 protein, the product of Cas9 mRNA in lonvo-z.

	Lonvo-z		Placebo	
	Antidrug antibodies (n=52)	**Anti-Cas9 antibodies (n=52)**	**Antidrug antibodies (n=28)**	**Anti-Cas9 antibodies (n=28)**
Baseline — no. (%)*				
Negative	51 (98)	52 (100)	26 (93)	28 (100)
Positive	1 (2)	0	2 (7)	0
Postbaseline — no. (%)†				
Negative	44 (85)	2 (4)	26 (93)	27 (96)
Positive	8 (15)	50 (96)	2 (7)	1 (4)

*Baseline is defined as the last measurement prior to the start of study intervention infusion.
†Postbaseline negative is defined as those who remained negative at all postinfusion time points. Postbaseline positive is defined as those who were positive at ≥1 postinfusion time point. Antidrug antibodies and anti-Cas9 antibodies were assessed pre-dose for day 1 infusion and crossover infusion and at weeks 2, 5, 12, 32, 56, 80, 104.